

# Grace Zhang

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## TECHNICAL SKILLS

**Languages:** Python, Go, TypeScript, JavaScript, Java, C/C++, SQL

**Frameworks & Libraries:** React, Django, Node.js, Express, Playwright, Electron, Tailwind CSS

**Tools & Platforms:** AWS, GCP, Docker, Git, Linux, PostgreSQL, MySQL

## EXPERIENCE

### Software Engineer Intern

Jan. 2026 – Apr. 2026

*Mantle Equity*

*Toronto, ON*

- Built a **Go** service that analyzed **Playwright** trace artifacts from **AWS S3** using **Gemini (Vertex AI)** to generate automated root cause reports, reducing debugging time from **20 minutes to seconds**
- Developed a full-stack investment holdings import/export workflow using **React**, **TypeScript**, **Go**, and **PostgreSQL**, streamlining bulk data updates through validated CSV/XLSX imports, reconciliation, and row-level error handling
- Delivered **5** production **TypeScript/Playwright** data ingestion workers for financial portals, improving pipeline reliability by resolving **10+** failures across MFA flows, dynamic tables, and document deduplication

### Software Developer Intern

May 2025 – Aug. 2025

*Empire Life*

*Kingston, ON*

- Built the withdrawals step of a high-visibility advisor wizard in **React/TypeScript** with bilingual validation, enabling accurate calculations and unblocking product testing
- Shipped advisor-facing features and production fixes for a **Django** web app used by thousands of advisors across Canada
- Reduced search latency by **40%** by integrating **Fuse.js** fuzzy search into a production **Django** application

### Web Developer Intern

Sept. 2024 – Dec. 2024

*University Health Network*

*Toronto, ON*

- Migrated the UHN Research website to **dotCMS**, creating reusable page templates to standardize layouts and streamline content publishing
- Engineered reusable UI components in **TypeScript** for the UHN Research intranet and public websites, improving responsiveness and maintainability
- Reduced manual content updates by **30%** by implementing dynamic content workflows using **dotCMS APIs** and **Velocity**

### Software Developer Intern

Jan. 2024 – Apr. 2024

*Pratt & Whitney*

*Mississauga, ON*

- Built a full-stack issue tracking module using **PHP**, **JavaScript**, and **SQL** to centralize and streamline requirements management workflows
- Implemented collapsible issue rows, nested comment threads, and file uploads, reducing issue resolution time by **20%**
- Developed backend scripts for binary file encoding, database storage, and retrieval, improving upload reliability and preventing data loss

### Robotics Operations & Software Developer Intern

May 2023 – Aug. 2023

*Swap Robotics*

*Kitchener, ON*

- Integrated **AWS SES** with **Express.js** to automate email alerts for safety infractions, reducing manual effort by **80%**
- Implemented backend services with **Node.js** and **Prisma ORM** for real-time safety monitoring and KPI reporting
- Revamped a Dockerized **React** dashboard with **REST API** integrations, automating safety tracking workflows for operations teams

## PROJECTS

### Overlayfy | *Electron, Node.js, Express, Spotify Web API, OAuth 2.0 PKCE*

[GitHub](#)

- Built a cross-platform **Electron** desktop app with a draggable overlay that displays real-time synchronized Spotify lyrics
- Implemented a **Node/Express** backend with **Spotify Web API** OAuth 2.0 PKCE authentication, LRCLib lyric retrieval, LRC timestamp parsing, and caching for repeated playback

## EDUCATION

### University of Waterloo

Waterloo, ON

*Bachelor of Computer Engineering*

*Expected May 2027*

- Relevant Courses:** Algorithms & Data Structures, Systems Programming, Database Systems, OS, Computer Networks